Water Security Plan Report

Date : 26/ 06/ 2023 Time : 12:08 PM User Name: Arkhi_155944

STATE: GUJARAT_24 » DISTRICT: BANASKANTHA_441 » BLOCK: DANTIWADA_3735 » GRAM PANCHAYAT: ARKHI_155944 » YEAR: 2023-24 »

Gram Panchayat Information

State:	Gujarat_24	District:	Banaskantha_441
Block / Taluk :	Dantiwada_3735	GP Name :	Arkhi_155944
GP ID :	155944	Block Area (Ha) :	45250.16
GP area (Ha) :	1094.83	Watershed Code :	
Watershed name :	BANAS RIVER	Watershed area in Block / Taluk (Ha) :	22400
Sub-basin Code :	20	Sub basin name :	0
Basin Code :	20	Basin Name :	Sabarmati

Location Map



Watershed Map

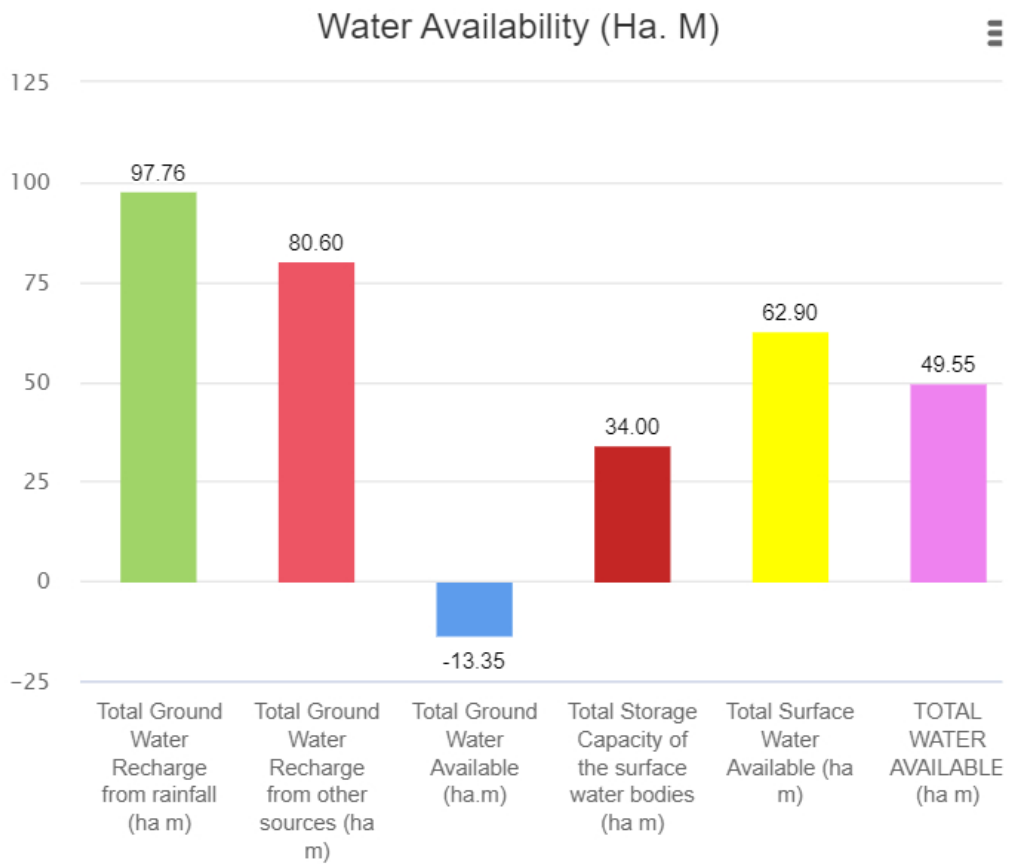

Note cam lite
Address : Vajapur,Gujarat,India
Latitude : 24.0771584°
Longitude : 72.77981462°
Altitude : 219.0 meter
Date : 02/19/2023 11:58 AM
Accuracy : 9.935046 meter
Time zone : GMT+05:30
Note : vajapur

Aquifer Map



Water Availability Report

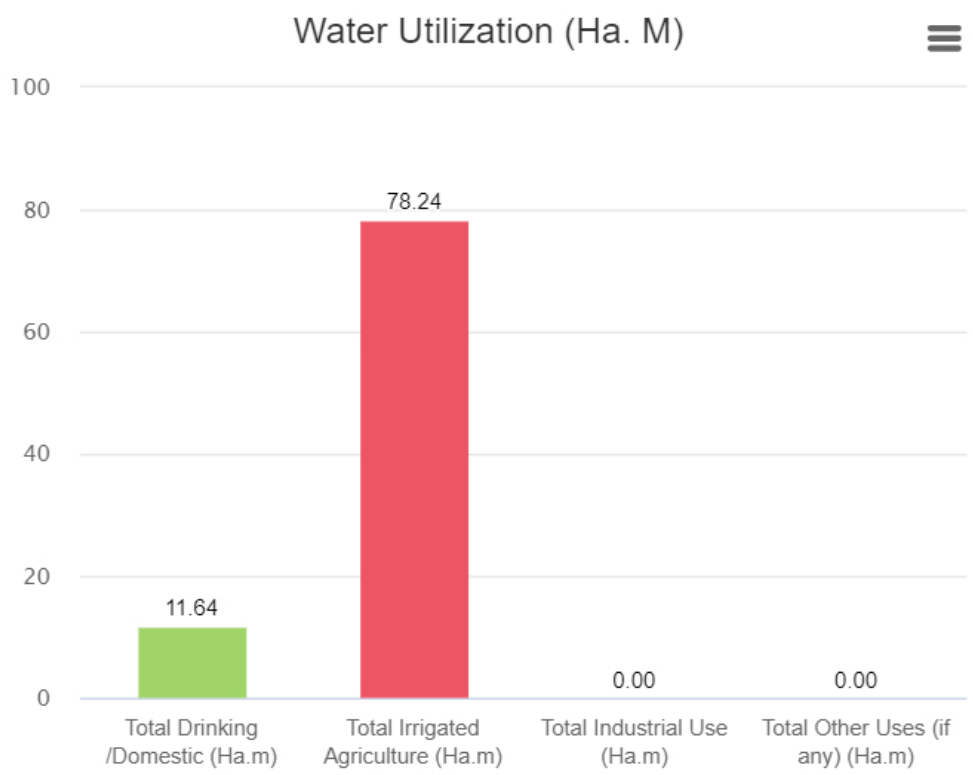
1.	Total Available Water From Rainfall	
a.	Annual Rainfall (mm)	494.96
b.	Estimated Volume of Rainfall (ha m)	541.90
c.	Evapotranspiration rate (mm)	348.33
d.	Estimated Evapotranspiration losses (ha m)	381.36
e.	Estimated Water Available From Rainfall (ha.m)	160.54
f.	Weighted Average Runoff Coefficient	0.10
g.	Rainfall Runoff(Ha.m)	16.05
2.	Ground Water	
a.	Recharge From Rainfall (ha.m)	
(i)	Monsoon	97.76
(ii)	Non-Monsoon	0.00
b.	Total Ground Water Recharge from rainfall (ha m)	97.76
c.	Ground Water Recharge from other sources (ha m)	
(i)	Monsoon	37.15
(ii)	Non-Monsoon	43.45
d.	Total Ground Water Recharge from other sources (ha m)	80.60
e.	Losses Due to Natural Discharge (ha.m)	191.71
f.	Total Ground Water Available (ha.m)	-13.35
g.	Ground Water Withdrawal (ha m)	17.84
h.	Categorization	Over-exploited



3.	Surface Water	
a.	Surface Water available in storages	
(i)	Number of Surface Water Bodies and Storage Capacity (ha m)	
Water Available in storages	Surface Water Bodies	Storage Capacity
Ponds	45	34
(ii)	Total Storage Capacity of the surface water bodies (ha m)	34.00
(iii)	Losses if Any (%)	15.00
(iV)	Surface Water Available in Storages (Ha.m)	28.90
b.	Transported Water from outside GP area (ha m)	34.00
c.	Total Surface Water Available (ha m)	62.90
4.	TOTAL WATER AVAILABLE	
	TOTAL WATER AVAILABLE (ha m)	49.55

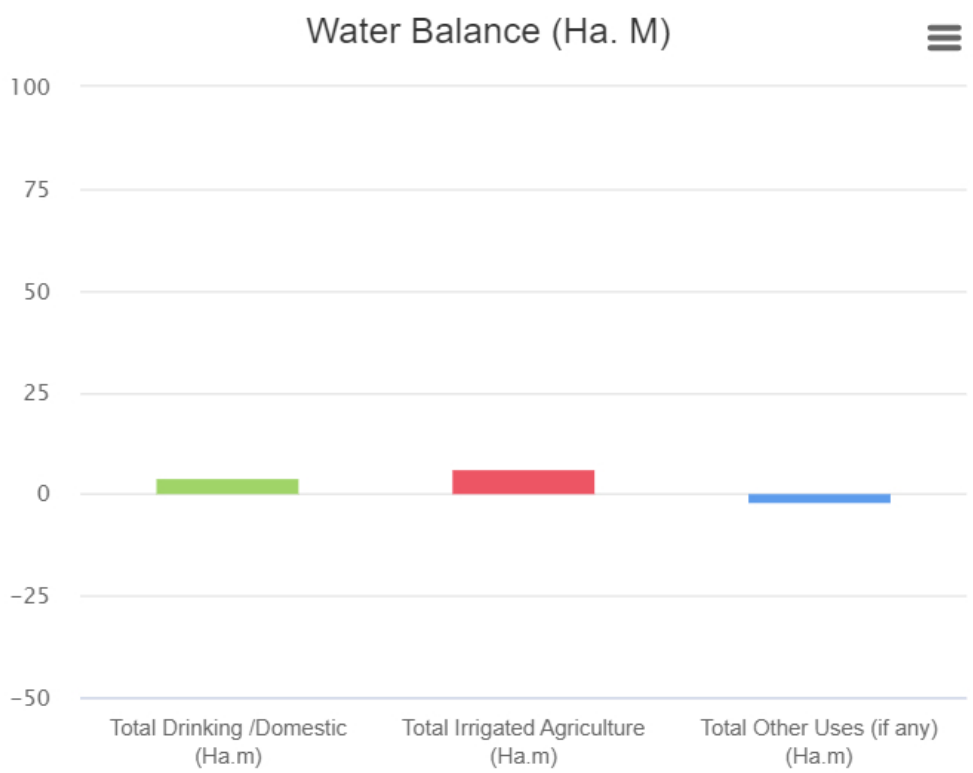
Water Utilization Report

Water Utilization		
A	Drinking /Domestic (ha m)	
i)	Form Surface Water	5.94
ii)	Form Ground Water	5.70
iii)	Drinking /Domestic (ha m)	11.64
B	Irrigated Agriculture (ha m)	
i)	Form Surface Water	34.43
ii)	Form Ground Water	43.81
iii)	Irrigated Agriculture (ha m)	78.24
C	Industrial Use (ha m)	
i)	Form Surface Water	0.00
ii)	Form Ground Water	0.00
iii)	Industrial Use (ha m)	0.00
D	Other Uses (if any) (ha m)	
i)	Form Surface Water	0.0001
ii)	Form Ground Water	0.0001
iii)	Other Uses (if any) (ha m)	0.0002



Water Balance Report

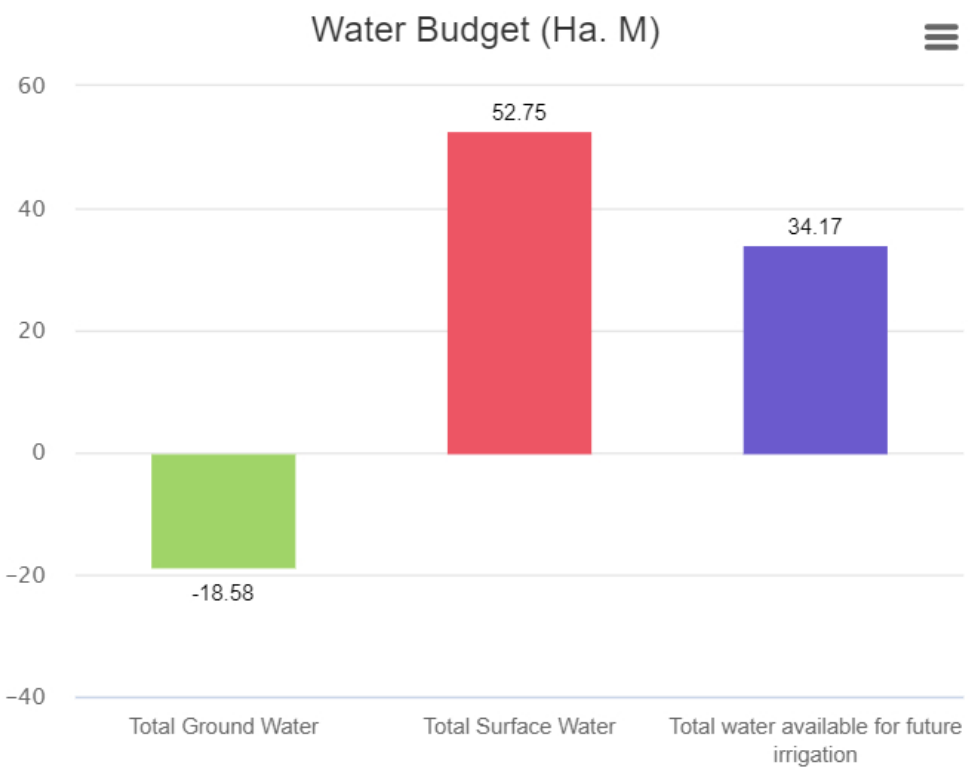
Water Balance		
A	Water Availability (ha m)	
i)	Total Ground Water Available	-13.35
ii)	Total Surface Water Available	62.90
iii)	Total Water Availability (ha m)	49.55
B	Water Utilization (ha m)	
i)	Total Surface Water Utilization	34.43
ii)	Total Ground Water Utilization	43.81
iii)	Total Water Utilization	78.24
C	Balance [Surplus (+) / Deficit (-)]	
i)	Total Ground Water Balance	-57.16
ii)	Total Surface Water Balance	28.47
iii)	Total Water Balance	-28.69



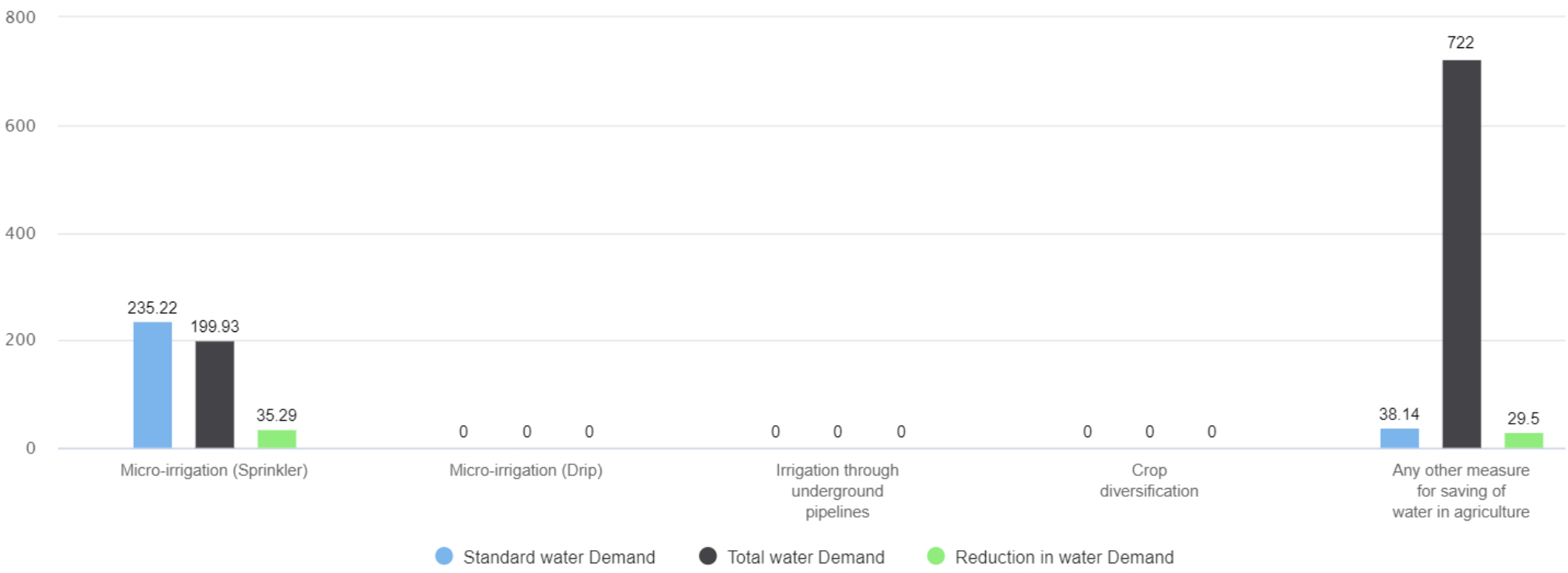
Water Budget Report

A	Allocation for Domestic & Industrial Requirements	
I	Humans	
i)	Present human population	3724.00
ii)	Annual population growth rate(r)	1.00
iii)	No. of years	28.00
iv)	Projected human population (as in 2050)	4920.00
v)	Per capita requirement (L)	55.00

vi)	Projected water requirement for humans (as in 2050) (ha m)	9.88
II	Livestock	
i)	Present livestock population	2046.00
ii)	Annual population growth rate(r)	1.00
iii)	No. of years	28.00
iv)	Projected human population (as in 2050)	2703.00
v)	Per capita requirement (L)	55.74
vi)	Projected water requirement for livestock (as in 2050) (ha m)	5.50
	Projected water requirement for humans and livestock (ha m) (as in 2050)	
	Total Projected water requirement for humans and livestock (as in 2050) (ha m)	15.38
A	Proposed share (in %) to be met from Ground water	34.00
B	Proposed share (in %) to be met from Surface water	66.00
C	Demand (in ha m) met from Ground water	5.23
D	Demand (in ha m) met from Surface water	10.15
C	Water available for future irrigation	
i	Ground water	-18.58
ii	Surface water	52.75
iii	Total water available for future irrigation	34.17



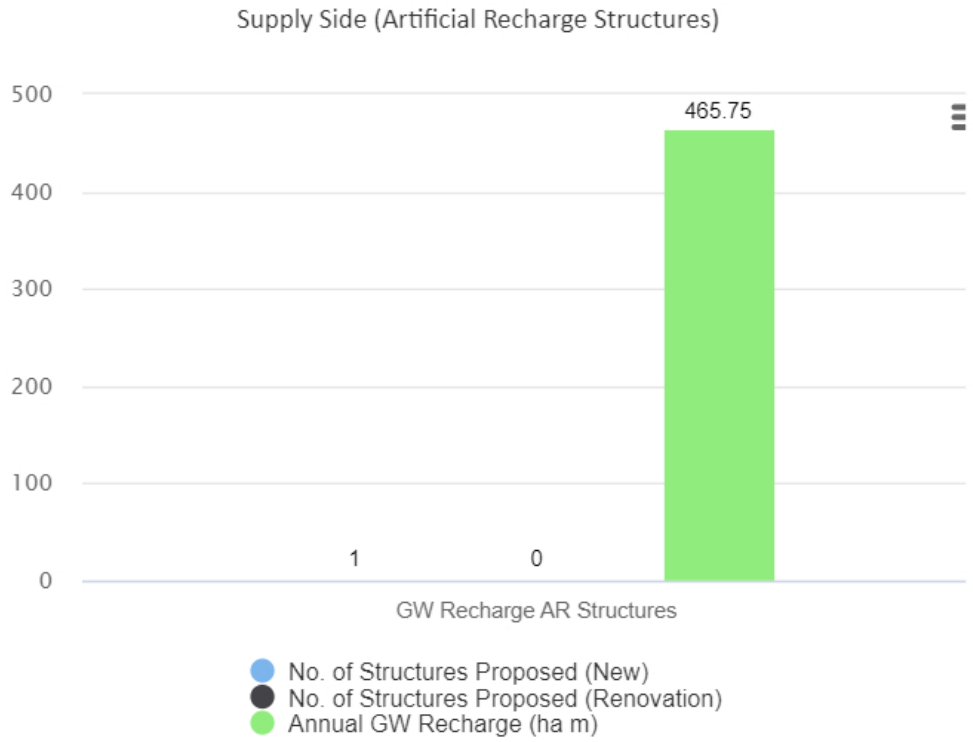
Demand Side

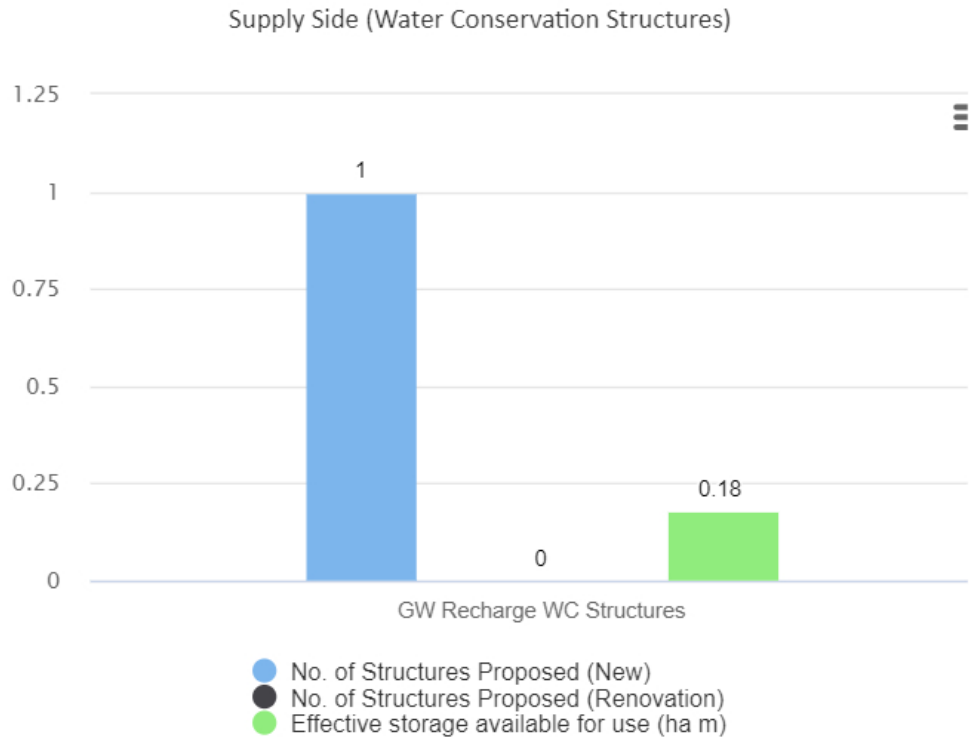


Demand Side							
Sr No	Name of Intervention	Area under Measures(ha)	Proposed Area(ha)	Total Water demand(ha m)	Net Water Demand(ha m)	Reduction in Water demand(ha m)	Approximate cost(in lakh)
1	Sprinkler	256.00	134.00	235.22	199.93	35.29	0.00
2	Drip	0.00	0.00	0.00	0.00	0.00	0.00
3	Pipelines	0.00	0.00	0.00	0.00	0.00	0.00
4	Crop	0.00	0.00	0.00	0.00	0.00	0.00
5	Any Other	0.00	398.00	38.14	722.00	29.50	0.00

Funding Source								
Sr No	Name of Intervention	Total Area under Conventional Irrigation (ha)	Total Area Proposed (ha)	Area Proposed (ha)	Financial Year	Department Name	Scheme Name	Approximate Cost(in Lakh)

Supply Side





Supply Side (Artificial Recharge Structures)										
SR No	Name of Structure	Storage Capacity(ha m)	Annual GW Recharge(ha m)	Village Name	Latitude	Longitude	Financial Year	Department Name	Scheme Name	Approximate Cost (in Lakhs)
1	Bore Hole Flooding(New)	345	465.75	Arkhi	76.5656	78.67	2023-24	Departmnet of Agriculture	IWDMP	1234567

Supply Side (Water Conservation Structures)										
SR No	Name of Structure	Storage Capacity(ha m)	Annual GW Recharge(ha m)	Village Name	Latitude	Longitude	Financial Year	Department Name	Scheme Name	Approximate Cost (in Lakhs)
1	Farm Pond(New)	34	0.18	Arkhi	45.	33.	2023-24	Rural Department and Panchayati Raj	Jalukt Shivar 2.0	1234567

PGWM Committee

Sr.no	Name	Designation	Gender	Mobile /Landline No.
1	Testing	asd	Transgender	

Sr.no	State	District	Block	Gram Panchayat	Date	GP Name	Village Name	Meeting Location	Male	Female	Transgender	Total number of attendees	MOM(Minutes of Meeting)	Photograph	Resolution
No Data															

MOM(Minutes of Meeting)

Photograph

Resolution

DPMU Declaration